

The Observer–Context Category: Relative Facts, a $\mathbb{Z}_2$ Switching Invariant, and the Movable Cut

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Abstract

We make “observer” a mathematical object: an isometric enlargement of the system together with a measurement context on the enlarged algebra, organized into a category (whose invertible switches form a subgroupoid) carrying the $U(1)$ cocycle of the companion papers. Three results follow, each backed by machine certificates and one anchored by superconducting-hardware data. (1) *Relative facts*: every observer admits an exact local classical section, while no global value assignment exists for the Peres–Mermin family — nor for any enlarged family containing its image under a Wigner enlargement. (2) A gauge-robust $\mathbb{Z}_2$ invariant of switch loops: $+1$ on all loops of Weyl switches, -1 on the oriented mutually-unbiased triangle. Nontriviality *requires* complementary switching; the converse fails, and the counterexample is pinned. The invariant counts τ -level switching parity — the same object as the Wigner–friend $-i$. (3) *The movable cut*: the comparison map between cut placements is fixed only up to section; its ray-level content is invariant (and measured), the section-level bookkeeping shifts by the loop holonomy, and the update rule available to each observer is itself forced. (4) The Frauchiger–Renner chain is *localized*: its observer cycle has trivial class and exactly invariant holonomy $+1$ — refuting the natural conjecture that such paradoxes measure holonomy — while its contradiction is state-sector, with contextual fraction exactly $1/6$ and the CHSH inequality at value $7/3$ as the unique separating facet of the codeword polytope. Observer-dependence, the triangle invariant, and cut conventionality are thereby organized by one cohomological object; we present them as structural characterizations within the stated axioms, not an interpretation-free derivation of Copenhagen.

1 Introduction

The Copenhagen principle of the companion note asserts that quantum theory is locally valid — classical within every measurement context — and globally obstructed. This paper extends that structure from contexts to *observers*. The blocking issue in earlier attempts was that “observer” remained informal. We repair this with a minimal definition sufficient for three theorems formalizing the relativity of facts, the holonomy content of complementary switching, and the movability of the Heisenberg cut. Every step is pinned by a machine certificate in the public verification suite (`observer_groupoid.py` and companions), and the holonomy split of Section 4 carries a hardware measurement.

2 The observer–context category

Fix the finite Weyl model at dimension d with m registers: observables are Weyl operators $W(v)$, a *context* is a closed commuting Weyl family (not necessarily maximal), and the section $W(u)W(v) = \tau^{c(u,v)}W(u+v)$, $\tau = e^{i\pi/d}$, defines the 2-cocycle c of the companion papers.

Definition 1 (Observer; morphisms). *An observer is a pair $o = (\iota, C)$: an isometric enlargement $\iota : \mathcal{H} \rightarrow \mathcal{H} \otimes \mathcal{H}_O$ (the cut datum) and a context C on the enlarged algebra. Morphisms are generated by (i) switches: unitaries of a fixed enlargement carrying C to C' , on which the Weyl-section gauge group acts by left and right multiplication; and (ii) refinements: further isometries $(\iota, C) \rightarrow (\iota' \circ \iota, C \otimes I)$ comparing cut placements. Observers and these morphisms generate the observer–context category $\mathbb{O}$: the switches are isomorphisms and form a subgroupoid, while the refinement isometries into larger Hilbert spaces are non-invertible, so $\mathbb{O}$ is a category (not a groupoid), and the holonomy invariants below are taken on the switch subgroupoid.*

Lemma 1 (Pullback and monotonicity). *A $\mapsto A \otimes I_O$ is an injective strict morphism of context structures pulling the cocycle back on the nose; in particular operator products of embedded contexts are unchanged, for any enlargement. Hence if a family admits no global value assignment, neither does any enlarged family containing its image: an assignment would restrict to one for the embedded copy. Certificate: the six embedded Peres–Mermin context products at $d = 8$ are $(+1)^{\times 5}, (-1)^{\times 1}$, identical to $d = 4$ (`observer_groupoid.py`, part A).*

3 Theorem 1: relative facts

Theorem 1. (a) *Every observer admits an exact local classical section: the cocycle restricted to any single context is a coboundary, with an explicit trivialization λ valued in μ_d for even d and μ_{2d} for odd d (sharp at $d = 3$); verified at $d = 2$ (all six PM contexts), $d = 3$, and $d = 4$ (`local_classicality.py`, `mu_d_valuation.py`). The earlier μ_{4d} / even- d half- τ -step wording is retracted (V46). (b) No global value assignment exists for the PM family, hence none for any enlarged family containing its image (PM product certificate plus Lemma 1).*

Each observer’s facts are classical; there is no observer-independent assignment of facts. Observer-dependence is the local-to-global cohomological gap — a theorem, not an interpretive gloss. The trivializations λ are the observer’s classical bookkeeping, and their mutual incompatibility around obstructed families is the entire content of contextuality at the observer level.

4 A $\mathbb{Z}_2$ switching invariant

For a loop of switches $U_k \cdots U_1$ closing on its initial context, define the section holonomy $\text{hol} = \det(W_c U_k \cdots U_1)$ with W_c any SU closer. The value hol itself is section-dependent (the lesson of the companion paper’s Wigner–friend analysis); its square is not:

Lemma 2 (Invariance — proved). *Every Weyl element $w = i^k P$ has $\det(w)^2 = 1$, and closers are special unitary. Hence Weyl regauging of any switch multiplies hol by ± 1 , and hol^2 is independent of regauging and of the closer. (Determinant fact machine-checked over all 16 Weyl elements.)*

Theorem 2. *At $d = 2$: (i) every loop of Weyl switches has $\text{hol}^2 = +1$; (ii) the oriented mutually-unbiased triangle $T \rightarrow T_x \rightarrow T_y \rightarrow T$ (switches H, S) has $\text{hol}^2 = -1$, with $\text{hol} = -i$ in the Clifford presentation. In that presentation $\text{hol}^2 = (-1)^{\#S \bmod 2}$: the invariant is the parity of τ -level switches. Certificates: 200 Weyl regaugings and 500 $SU(2)$ closers, single values throughout.*

Corollary 1. *$\text{hol}^2 = -1$ requires complementary switching: loops confined to Weyl switches (in particular, context-preserving ones) are trivial.*

Remark (The converse fails — pinned). *The 2-cycle $T \rightarrow T_x \rightarrow T$ with switches (H, H) traverses a complementary pair yet has $\text{hol}^2 = +1$ (machine-pinned). The invariant therefore detects the oriented MUB triangle — the triality that produces the Wigner–friend $-i$ — not pairwise complementarity. We state this openly; “complementarity is holonomy” in the equivalence sense is false, and necessity (Corollary 1) is the correct claim.*

Hardware: on `ibm_fez` the ray-level holonomy of the triangle loop measures $+1$ (loop phase $-2.9^\circ \pm 1.6^\circ$), excluding $-i$ at ray level by $\sim 56\sigma$ (shot-noise statistical, under the implemented interferometric model) — exactly the section split the theorem requires.

Conjecture 1. *In the Clifford presentation at general d , $\text{hol} \in \mu_{2d}$ and the Weyl-gauge-invariant residue $\text{hol}^2 \in \mu_d$ detects the top layer of the 2-adic tower of the companion papers.*

5 Theorem 3: the movable cut

Theorem 3. *Let two observers differ only by a refinement (friend versus Wigner descriptions of one run). The comparison map between their descriptions is fixed only up to the section choice: its ray-level content is invariant — and is what the hardware measures — while the section-level bookkeeping shifts by the loop holonomy of Section 4. Three levels must be kept apart: the empirical probabilities are invariant under the move; the determinant-section phase shifts by the loop holonomy; and the cohomology class is unchanged. In this precise sense ray-level physics is independent of the cut placement while the section bookkeeping records the move; the broader reading “the cut may be moved without empirical consequence” is the interpretation this supports, not a claim beyond the three-level statement. Moreover the update rule available to each observer is constrained (not a free convention): statistics, repeatability, and commutant covariance force the one-parameter affine depolarizing family per degenerate block (complete-positivity interval $-1/(r^2 - 1) \leq p \leq 1$), with the Lüders instrument the unique single-Kraus member (`lueders_instrument.py`, a numerical stress test; CP interval certified in `lueders_cp_interval.py`).*

This is consistent with Copenhagen’s conventionality of the cut, now given structural content: the cut is a choice of trivialization, ray-level physics is invariant under the choice, and the invariant that survives is precisely the class. We offer this as a structural reading, not a claim that it is the only interpretation the formalism admits.

6 Theorem 4: the Frauchiger–Renner cycle, localized

Model each lab by one qubit (memory identified with the measured system, as is standard); the probabilistic core of the FR chain is Hardy’s structure on $\psi = (|h, \downarrow\rangle + |t, \downarrow\rangle + |t, \uparrow\rangle)/\sqrt{3}$

with the four observer contexts $\{\bar{Z}Z\}, \{\bar{Z}X\}, \{\bar{X}X\}, \{\bar{X}Z\}$ — a closed 4-cycle Weyl family in $\mathbb{O}$ (`fr_cycle.py`).

Theorem 4. (i) *The FR observer cycle has trivial class (all four context phases +1; the value system is solvable, $|L| = 16$) and its switch loop closes exactly with $\text{hol} = +1$; since every two-qubit Weyl element has determinant +1, this holonomy is fully gauge-invariant. The FR paradox is therefore not a holonomy phenomenon: the natural conjecture that observer-cycle paradoxes measure holonomy is refuted by the machinery of this program itself.* (ii) *The paradox is state-sector: $\text{CF}(F_{\text{FR}}, \psi) = 1/6$ exactly, and by the codeword-polytope theorem of the companion paper the FR contradiction is equivalent to the moment vector of ψ lying outside $\text{conv } L$. The separating facet, extracted by exact separation, is the CHSH inequality $\langle \bar{Z}Z \rangle - \langle \bar{Z}X \rangle + \langle \bar{X}X \rangle + \langle \bar{X}Z \rangle \leq 2$ with quantum value $7/3$, and the Abramsky–Barbosa–Mansfield normalization reproduces the LP value: $\text{CF} = (7/3 - 2)/2 = 1/6$.*

The Hardy reading of FR is known; the contribution here is its exact placement in the program’s taxonomy. Consistency conditions for observer networks are polytope-membership conditions in the state sector, not holonomy conditions: “absoluteness of observed events” fails where the codeword polytope is exited, at a quantified rate.

7 Discussion and open problems

Open: the general- d tower refinement of Section 4; multi-round dynamics beyond a single update; whether any observer-network paradox is genuinely holonomy-sector (Theorem 4 says FR is not — a candidate would need an odd cycle, i.e. an AvN core); and review of this draft beyond its adversarial self-passes.

References

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